Assignment 6

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Econometrics I
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Instructions. Two problems on regression discontinuity and synthetic control (Sessions 10–11). `R` with `fixest` or Python with `pyfixest`. For the synthetic control you may use a package (`augsynth` recommended, installed from GitHub repository `ebenmichael/augsynth` with `remotes::install_github`; `tidysynth` or `synthdid` also fine; `pysyncon` in Python) or write the eight-line version described in Problem 2. Turn in a PDF with your results and reasoning, and your code.

1 Rating rounding: sharp, then fuzzy

Based on Anderson and Magruder (2012), who use Yelp's half-star rounding. A platform computes each restaurant's average rating r_i to two decimals but *displays* it rounded to the nearest half star. Restaurants with $r_i \geq 3.75$ display four stars; those just below display three and a half. Simulate 3,000 restaurants with $r_i \sim \text{Uniform}(3.25, 4.25)$, let $x_i = r_i - 3.75$ be the distance to the threshold, and let log weekly revenue be

$$y_i = 10 + 1.5\,x_i + 0.8\,x_i^2 + 0.6 \cdot \mathbf{1}\{x_i \geq 0\} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, 1).$$

Underlying quality matters smoothly; the displayed star adds a jump of 0.6 at the threshold.

(a) Naive comparison: mean y for restaurants displaying four stars minus those displaying three and a half. Why is this not the effect of the extra half star?

(b) Binscatter: mean y in 20 equal-width bins of x; plot with a vertical line at zero.

(c) Local linear RDD: regress y on $\mathbf{1}\{x \geq 0\}$, x, and their interaction, using restaurants within h of the threshold, for $h = 0.5$, 0.25, and 0.10. Report the jump and its robust standard error at each bandwidth. What is the tradeoff as h shrinks, and how does the $0.8\,x^2$ term in the truth show up in it?

(d) Placebo cutoffs at $x = -0.25$ and $x = +0.25$, same bandwidth 0.25. What should you find, and what do you find?

(e) Manipulation check: compare the number of restaurants just below and just above the threshold (within 0.05). Explain in one paragraph what a pile-up just above would suggest in the real setting, and why it would invalidate the design.

(f) Fuzzy version. Now the displayed star does not move revenue directly; instead it raises the chance the platform *features* the restaurant. Simulate an unobserved quality $q_i \sim \mathcal{N}(0,1)$ and

$$F_i = \mathbf{1}\{-0.8 + 1.4 \cdot \mathbf{1}\{x_i \geq 0\} + 0.8\, q_i + \nu_i > 0\}, \quad y_i = 10 + 1.5\, x_i + 0.8\, x_i^2 + 1.0\, F_i + 0.8\, q_i + \varepsilon_i,$$

with $\nu_i, \varepsilon_i \sim \mathcal{N}(0,1)$. The effect of being featured is 1.0. Within $h = 0.25$: (i) regress y on F (and the running-variable terms); (ii) estimate the reduced form and the first stage at the threshold; (iii) form the Wald ratio and compare to 2SLS with the threshold dummy as the instrument for F. Why does (i) fail and (iii) work, and for which restaurants is the 2SLS number the effect of being featured?

2 A soda tax, one city, thirty donors

A city imposes a soda tax at the start of quarter 13 of a 20-quarter panel; 30 other cities do not. Per-capita soda sales follow a factor structure,

$$Y_{jt} = 100 + 5\,(\lambda_{1t}\,\mu_{j1} + \lambda_{2t}\,\mu_{j2}) + \varepsilon_{jt}, \qquad \varepsilon_{jt} \sim \mathcal{N}(0,1),$$

with λ_{1t} a linear trend from 0 to 2 over the 20 quarters, $\lambda_{2t} \sim \mathcal{N}(0,1)$ i.i.d. over t, and city loadings $\mu_{j1}, \mu_{j2} \sim \text{Uniform}(0,1)$ for the donors. Give the treated city loadings $(0.9, 0.5)$: it is a fast-growing city. The tax cuts its sales by 10% in every post-tax quarter (multiply Y_{1t} by 0.9 for $t \geq 13$).

(a) Simulate. Plot the treated city against the average of the donors. Estimate the effect by difference-in-differences with all donors as the control group. Why is it biased here, and in which direction?

(b) Build the synthetic control: donor weights $w \geq 0$, $\sum_j w_j = 1$, chosen to minimize the pre-period squared gap $\sum_{t \leq 12} \left(Y_{1t} - \sum_j w_j Y_{jt}\right)^2$. (Eight-line version: parameterize $w_j = e^{\theta_j} / \sum_k e^{\theta_k}$ and hand the objective to `optim`; or use `quadprog::solve.QP`; or a package.) Report the weights above 0.05, the pre-period RMSPE, and the average post-period gap. Compare to the truth and to (a).

(c) Why does the synthetic control get this right when DiD does not? Answer in terms of the factor loadings.

(d) Placebo in space: give the treatment to each donor in turn (with the treated city removed from the donor pool), compute each donor's ratio of post- to pre-period RMSPE, and place the treated city's ratio in that distribution. Report the implied p-value. What does this test assume?

(e) Conformal inference (Chernozhukov, Wüthrich and Zhu 2021; Session 11 slides). For a candidate effect θ: subtract θ from the treated city's sales in quarters 13–20, refit the weights on *all* 20 quarters, and take the residuals $\hat{u}_t$. Let S be the mean of $|\hat{u}_t|$ over quarters 13–20, recompute S for each of the 20 cyclic shifts of $(\hat{u}_1, \ldots, \hat{u}_{20})$, and let $p(\theta)$ be the share of shifts with S at least as large as the unshifted one. (`augsynth`'s `summary(fit, inf_type = "conformal")` does a per-quarter version; the full test is about ten lines by hand.)

(i) Report $p(0)$ and the 90% confidence set $\{\theta : p(\theta) > 0.10\}$ on a grid of θ.

(ii) Keep only the first 8 donors and redo the placebo test in (d) and $p(0)$. What is the smallest p-value each test can produce, and what does that mean for a city with few comparable donors?

(f) Now move the treated city's loadings to $(1.3, 0.5)$, outside the range of the donors. Re-run (b). What happens to the pre-period fit, and what does that tell you about when synthetic control can and cannot be used?